Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 01)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

0 1 . 2 Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

Useful properties

Properties causing ozone depletion

0 1 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

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0 1 . 4 State what is meant by an **ozone hole**.

[1 mark]

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0 1 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

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0 1 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

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0 2 . 1 Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

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0 2 . 2 Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

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0 2 . 3 Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

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0 2 . 4 Give **three** other positive feedback mechanisms.

[3 marks]

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0 2 . 5 Explain what is meant by a climate **tipping point**.

[2 marks]

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0 3 . 1 Which of the following best describes the purpose of Critical Group Monitoring? Tick one box.

[1 mark]

A To detect health damage in the population most affected by a pollutant ☐

B To predict the route a pollutant will take through the environment ☐

C To assess exposure of the people most at risk so that everyone else can be assumed to be sa ☐

D To measure the concentration of a pollutant at the point of release ☐

0 3 . 2 Explain why CPA can reduce the cost of an environmental monitoring programme.

[2 marks]

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0 3 . 3 A new chemical has properties similar to PCBs (persistent, liposoluble chlorinated organic compounds). Explain how knowledge of PCBs helps decisions about the new chemical.

[2 marks]

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0 4 . 1 Explain how occupancy sensors and programmable thermostats reduce energy waste.

[2 marks]

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0 4 . 2 Explain why LED lights use much less electricity than the filament and CFL lamps they replaced, and give **two** other examples of low-energy appliance design.

[3 marks]

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0 4 . 3 Describe **three** changes in human behaviour that reduce domestic energy use.

[3 marks]

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0 4 . 4 Explain why heating water on demand saves energy compared with storing hot water.

[2 marks]

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0 5 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 5 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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0 5 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

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0 6 . 1 Describe how DNA databases are used to control the illegal trade in tropical timber and ivory.

[3 marks]

0 6 . 2 Explain the difference between radio / GPS tracking and acoustic tracking and state when each is used.

[2 marks]

0 6 . 3 Give **three** types of carrier system used to move sensors to study areas and one advantage of drones for ecological monitoring.

[3 marks]

0 6 . 4 Explain how motion-sensitive cameras and CCTV reduce disturbance to wildlife compared with direct observation.

[2 marks]

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0 7 . 1 Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

0 7 . 2 Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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0 7 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 8 . 1 Describe how human activities have altered the natural equilibria of the carbon and nitrogen cycles, and evaluate the strategies that could counteract these changes.

[9 marks]

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0 8 . 2 State what is meant by the **residence time** of a substance in a reservoir.

[1 mark]

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0 9 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 9 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 9 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 9 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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1 0 Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O_2)	2
Triatomic oxygen / ozone (O_3)	3

1 0 . 1 Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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1 0 . 2 Give **two** other ways in which the atmosphere supports life.

[2 marks]

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1 0 . 3 Explain why the atmosphere is thermally stratified into layers.

[1 mark]

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1 0 . 4 The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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Write an essay on one of the following topics.

11 . 1 Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

OR

11 . 2 Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS